

Ejercicio 6: Simplificación mediante el álgebra de Boole

6.1. Mediante las técnicas del álgebra de Boole, simplificar las siguientes expresiones lo máximo posible:

- (a) $A(A+B)$ (b) $A(\bar{A}+AB)$ (c) $BC+\bar{B}C$
 (d) $A(A+\bar{A}B)$ (e) $A\bar{B}C+\bar{A}BC+\bar{A}\bar{B}C$

$$\begin{aligned} \text{e) } & A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C = \\ & = (\bar{A}\bar{B} + \bar{A}B + A\bar{B})C \\ & = (\bar{A}\bar{B} + (\bar{B} + B)\bar{A})C \\ & = (\bar{A}\bar{B} + 1 \cdot \bar{A})C \\ & = (\bar{A}\bar{B} + \bar{A})C \\ & = (\bar{A} + A\bar{B})C \\ & = (\bar{A} + \bar{B})C \end{aligned}$$

Teorema 14: $a + (\bar{a} \cdot b) = a + b$

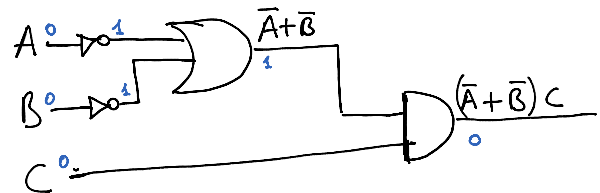
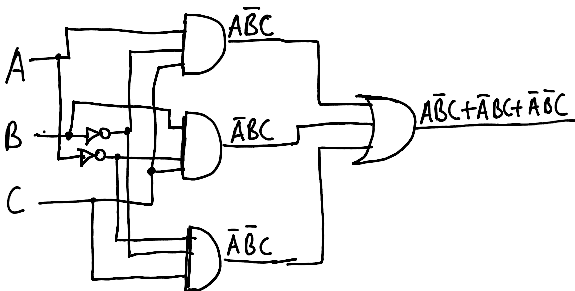
Teorema 15: $a \cdot (\bar{a} + b) = a \cdot b$

• Cantidad de filas de la tabla = 2^n , donde n es la cant. de variables

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$A\bar{B}C$	$\bar{A}BC$	$\bar{A}\bar{B}C$	$A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$	$\bar{A} + \bar{B}$	$(\bar{A} + \bar{B})C$
0	0	0	1	1	1	0	0	0	0	1	0
0	0	1	1	1	0	0	0	1	1	1	1
0	1	0	1	0	1	0	0	0	0	1	0
0	1	1	1	0	0	0	1	0	1	1	1
1	0	0	0	1	1	0	0	0	0	1	0
1	0	1	0	1	0	1	0	0	1	1	1
1	1	0	0	0	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0	0

$$A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$$

$$(\bar{A} + \bar{B}) \cdot C$$



6.2. Mediante las técnicas del álgebra de Boole, simplificar las siguientes expresiones:

- (a) $(A + \bar{B})(A + C)$ (b) $\bar{A}B + \bar{A}B\bar{C} + \bar{A}BCD + \bar{A}B\bar{C}\bar{D}E$
 (c) $AB + \bar{A}BC + A$ (d) $(A + \bar{A})(AB + ABC\bar{C})$
 (e) $AB + (\bar{A} + \bar{B})C + AB$

$$\text{a) } (A + \bar{B})(A + C) = A \cdot A + A \cdot C + \bar{B} \cdot A + \bar{B} \cdot C = A + AC + A\bar{B} + \bar{B}C$$

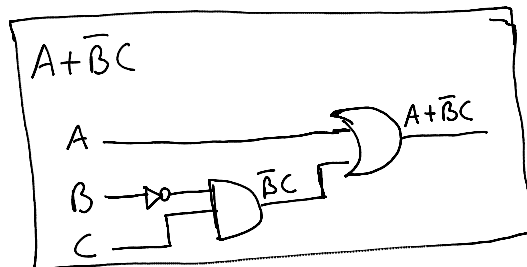
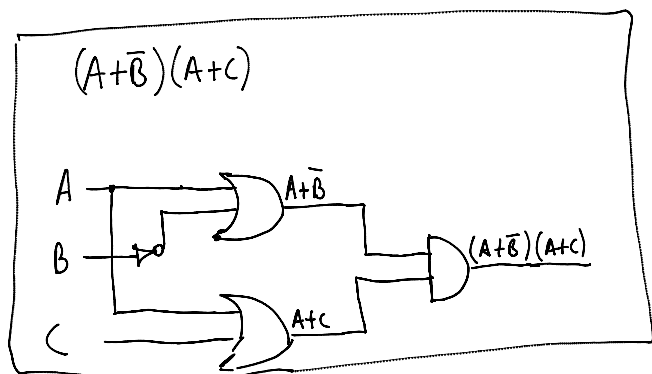
Teorema 6: $a + a \cdot b = a$

$$\begin{aligned}
 a) (A+\bar{B})(A+C) &= AA+AC+A\bar{B}+\bar{B}C \\
 &= \underbrace{A+AC+A\bar{B}+\bar{B}C}_{A+A\bar{B}+\bar{B}C} \\
 &= \boxed{A+\bar{B}C}
 \end{aligned}$$

Teorema 6: $a + a.b = a$

A	B	C	$\bar{B}$	$A+\bar{B}$	$A+C$	$(A+\bar{B})(A+C)$	$\bar{B}C$	$A+\bar{B}C$
0	0	0	1	1	0	0	0	0
0	0	1	1	1	1	1	1	1
0	1	0	0	0	0	0	0	0
0	1	1	0	0	1	0	0	0
1	0	0	1	1	1	1	0	1
1	0	1	1	1	1	1	1	1
1	1	0	0	1	1	1	0	1
1	1	1	0	1	1	1	1	1

Como son iguales los resultados segun las entradas entonces la expresion algebraica reducida es equivalente a la original.



$$\begin{aligned}
 (b) \quad &\bar{A}B + \bar{A}B\bar{C} + \bar{A}BCD + \bar{A}B\bar{C}\bar{D}E \\
 &= \bar{A}B + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D}E \\
 &= \bar{A}B + \bar{A}B\bar{C}\bar{D}E \\
 &= \bar{A}B
 \end{aligned}$$

32 filas

Teorema 6: $a + a.b = a$

Teorema 14: $a + \bar{a}.b = a + b$

Teorema 2: $a + a = a$

Teorema 6:

$$\begin{aligned}
 a=0, b=0 &\} 0+0.0=0 \\
 a=0, b=1 &\} 0+0.1=0 \\
 a=1, b=0 &\} 1+1.0=1 \\
 a=1, b=1 &\} 1+1.1=1
 \end{aligned}$$